Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building and maintaining **mobile and consumer applications**, cross-platform services, and synchronized workflows using **Java**, Android, Python, React, REST APIs, testing, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*
- Architected a multi-tenant platform using Azure SQL and Blob Storage with row-level isolation and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built API integrations across FastAPI, PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, error handling, and CI/CD releases sustaining 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*
- Built responsive React interfaces and Python services for uploads, history, configuration, and live feedback, reducing user turnaround 60% while supporting reliable production workflows used by 150+ instructors.
- Designed concurrent processing and secure multi-format ingestion through REST APIs and AWS RDS, executing 1K+ files per batch with fault-tolerant recovery, usage tracking, and stable performance under load.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*
- Engineered cross-platform software with microservices and CI/CD pipelines for unit and integration testing across Windows, Linux, and macOS while collaborating with Microsoft and GitHub stakeholders in Agile sprints.
- Automated nightly builds, executable installation, integration tests, unit tests, and reports through Jenkins, validating reliable cross-platform software synchronization on virtual machines while reducing manual entry by 35%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*
- Implemented eight modular services for authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across production workflows.
- Built responsive React workflows and automated Docker deployments through Jenkins CI/CD, achieving 95%+ build validation while integrating operational monitoring and analytics across controlled staging and production environments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Built cross-platform mobile and web booking workflows with Flask APIs, Firebase synchronization, payments, and webhooks, processing hourly Databricks micro-batches across 50K+ records with consistently reliable sub-200-millisecond responses.
- Designed end-to-end workflows, mapped 12+ integration points, and presented technical solutions to client stakeholders, reducing design errors 25% while supporting effective collaborative cross-functional implementation planning and alignment.

Skills

Mobile & Languages: Java, Android, C++, Python, JavaScript, TypeScript, SQL, HTML, CSS, React, Responsive Design
Application Development: OOP, REST APIs, WebSockets, Authentication, Authorization, Cross-Platform Development, Data Synchronization, PostgreSQL, SQL Server, MySQL, MongoDB, Firebase
Testing & Diagnostics: Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Debugging, Defect Resolution, Error Handling, Monitoring, Sentry, Performance Optimization
Delivery: Git, GitHub, Jenkins, GitHub Actions, CI/CD, Docker, AWS, Azure, Agile/Scrum, Code Reviews, Documentation, Production Support

Projects

Chat2SQL Code | *Java, TypeScript, Docker, Windows, Linux, macOS*
- Worked across Java and TypeScript modules using Docker, Maven, Node.js, and Yarn to support tested cross-platform database workflows for 10+ systems on Windows, macOS, and Linux environments.

HMIS - Hospital Management Information System Live | *TypeScript, MERN, REST APIs, Sentry*
- Implemented consent-based access, mobile-responsive workflows, REST integrations, automated alerts, and Sentry monitoring through scalable TypeScript services supporting 100+ daily operations while reducing hospital administrative workload by 30%.

CPU-scheduling Simulator Live | *C++, React, Operating Systems, Algorithm Verification*
- Implemented six C++ scheduling algorithms with edge-case handling, interactive controls, Gantt visualization, and dynamic performance metrics, systematically testing functional correctness across 1K+ preemptive and non-preemptive process combinations.